

TODOS — Nano-Rare GT Framework

M1: PoC Pipeline (Week 1-2) — IN PROGRESS

- ☐ Project setup + architecture (ADR-001)
- ☐ Disease discovery (Orphanet + OMIM)
- ☐ Protein linkage (UniProt + AlphaFold)
- ☐ Vector sizing + serotype mapping
- ☐ CLI end-to-end match
- ☐ v0.1 tag + review

M2: Multi-Disease Batch (Week 3-4)

- ☐ Batch query parallelization
- ☐ Full Orphanet corpus query
- ☐ SQL filtering + progress bars
- ☐ v0.2 tag

M3: Full 9-Dimension Scoring (Week 5-6)

- ☐ Immunogenicity (IEDB)
- ☐ Promoter matching (TiProD/GEO)
- ☐ Codon optimization (CAI)
- ☐ Therapeutic window / toxicity
- ☐ Cross-correction potential
- ☐ Weight calibration + sensitivity analysis
- ☐ v0.3 tag

M4: Regulatory Brief Generator (Week 7-8)

- ☐ Jinja2 templates for Precedent Brief
- ☐ FDA CBER / EMA CAT guidance sections
- ☐ CMC feasibility flags
- ☐ v0.4 tag

M5: Validation (Week 9-10)

- ☐ Retrospective case study #1
- ☐ Retrospective case study #2 (failure mode)
- ☐ Weight tuning from validation
- ☐ v0.5 tag

M6: Public Release (Week 11-12)

- ☐ Open-source on GitHub (MIT)
- ☐ Documentation site
- ☐ Preprint submission
- ☐ Outreach to patient orgs + biotech BD
- ☐ v1.0 tag

Deferred / Backlog

- ☐ PostgreSQL backend (v2)
- ☐ FastAPI service layer (v2)
- ☐ React dashboard (v3)
- ☐ Real-time ClinicalTrials.gov sync
- ☐ Manufacturing cost modeling (COGS)
- ☐ Patient foundation fundraising impact analysis

☐ FDA dialog simulation module